

Week 21: Project Work – Planning & Setup

1. Project Idea Finalization

The first step in project development is selecting and finalizing a practical project idea. The project should solve a real-world problem and align with MERN stack technologies. Proper planning helps define features, scope, and project goals clearly. This stage is important for creating a strong development roadmap.

2. MongoDB Schema Design

Schema design is essential for organizing database structure efficiently. Collections, fields, and relationships between data models are planned before development starts. A well-designed MongoDB schema improves data consistency, scalability, and performance.

3. React Component Planning

Before frontend implementation, the application UI is divided into reusable React components. Pages such as login, dashboard, profile, and forms are planned in a structured way. This improves code maintainability and reusability.

4. Git Branching Strategy

Git branching helps manage project development efficiently in teams or large projects. Separate branches like main, development, and feature branches are used for better code organization. This reduces conflicts and improves version control management.

Conclusion

This week focused on project planning and setup. By finalizing the project idea, designing schemas, planning React components, and using Git branching, a strong development foundation was prepared.